TUTORIAL 2

Proof Methods, Sets, and Fields

MOTIVATION AND GUIDING IDEA

A proof must work within the given setting: equality of sets is proved by two inclusions, and a field identity is proved from the field axioms. In a finite field, addition, subtraction, multiplication, and division by nonzero elements are permitted; numerical notation does not imply an ordering like that of $\mathbb{R}$.

NOTATION AND CONVENTIONS

The symbol $x \in A$ expresses membership of an element, whereas $A \subseteq B$ expresses inclusion of sets. The quantifiers $\forall$ and $\exists$ mean ``for every'' and ``there exists''; their order matters. The set $A \setminus B$ consists of elements of A absent from B, and $|A|$ is its cardinality when A is finite. For a field F, write $F^\times = F \setminus \{0\}$. The notation $a \equiv b \pmod{p}$ means $p \mid (a-b)$. For prime p, writing $\mathbb{Z}_p = \{0, \ldots, p-1\}$ uses representatives of residue classes. The symbol $\mathbb{F}_q$ denotes a field with q elements; when $q = p^n$ with $n > 1$, it must not be identified with arithmetic modulo q.

DEFINITION 2.1

Equality of sets and cancellation. For sets A, B, we have $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$. To prove equality, take an arbitrary element from each side and prove that it belongs to the other side. In a field F, for $a, b, c \in F$ with $c \neq 0$,

$$ac = bc \quad \Longleftrightarrow \quad a = b,$$

because multiplication by c^{-1} is permitted. The condition $c \neq 0$ is essential. The characteristic of a field is the least positive integer p for which $p \cdot 1 = 0$, if such an integer exists; otherwise its characteristic is 0. The characteristic of a finite field is prime.

MOTIVATION AND GUIDING IDEA

Objectives. We will strengthen proofs by induction and proofs of equivalence, identify the field axioms, and calculate in finite fields. We will distinguish familiar real arithmetic from arithmetic adapted to the specified field.

The first part is a focused review of proof methods and the language of sets. The second part introduces fields and finite field arithmetic; the order used in class will follow the progress of the lectures.

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2.1 The induction principle

DEFINITION 2.2

The principle of mathematical (weak) induction:
Let n_0 be a nonnegative integer, and let $P(n)$ be a statement defined for every integer $n \geq n_0$. Suppose that:

1. **Base case:** $P(n_0)$ is true.
2. **Inductive step:** for every integer $n \geq n_0$, the assumption $P(n)$ implies $P(n+1)$.

Then $P(n)$ is true for every integer $n \geq n_0$.

EXERCISE 2.1

Please prove by induction that, for every $n \geq 1$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

WORKED SOLUTION

The statement: $P(n) : \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.
Base case: for $n = 1$, the left side is 1, and the right side is $\frac{1 \cdot 2}{2} = 1$.
Induction hypothesis: suppose $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$ for some $k \geq 1$.
Inductive step:

$$\sum_{i=1}^{k+1} i = \left(\sum_{i=1}^{k} i\right) + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}.$$

This is the formula for $k+1$.
Conclusion: the statement holds for every $n \geq 1$.

EXERCISE 2.2

Please prove by induction that $13 \mid (14^n - 1)$ for every integer $n \geq 1$.

WORKED SOLUTION

For the base case $n = 1$, $14 - 1 = 13$ is divisible by 13. Suppose that, for some $k \geq 1$, there is $m \in \mathbb{Z}$ such that $14^k - 1 = 13m$. Then

$$14^{k+1} - 1 = 14(14^k - 1) + 13 = 14 \cdot 13m + 13 = 13(14m + 1).$$

This expression is divisible by 13, completing the inductive step. Since the base case and inductive step hold, the claim is true for every $n \geq 1$. For an additional check, $14 \equiv 1 \pmod{13}$, so $14^n \equiv 1^n = 1 \pmod{13}$.

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EXERCISE 2.3

Please prove that, for every $n \geq 4$,

$$2^n \geq n^2.$$

WORKED SOLUTION

We use induction on n.
Base case: for $n = 4$, $2^4 = 16 = 4^2$.
Induction hypothesis: suppose $2^k \geq k^2$ for some $k \geq 4$.
Inductive step:

$$2^{k+1} = 2 \cdot 2^k \;\geq\; 2k^2.$$

It suffices to show $2k^2 \geq (k+1)^2 \iff k^2 - 2k - 1 \geq 0$. For $k \geq 4$, $k^2 - 2k - 1 > 0$, so $2^{k+1} \geq (k+1)^2$.
By induction, $2^n \geq n^2$ for every $n \geq 4$.

2.2 Sets

DEFINITION 2.3

- **Inclusion:** $A \subseteq B$ means that every element of A also belongs to B.
- **Union:** $A \cup B = \{\, x \mid x \in A \text{ or } x \in B \,\}$.
- **Intersection:** $A \cap B = \{\, x \mid x \in A \text{ and } x \in B \,\}$.

$$\Omega = \{1,2,3,4,5\}, \quad A = \{1,2,3\}, \quad B = \{2,3,4\}$$

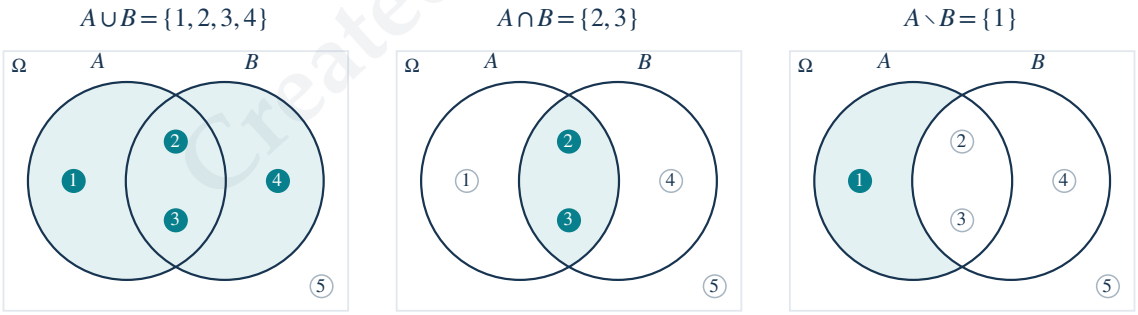


Figure 2.1. For $A = \{1,2,3\}$ and $B = \{2,3,4\}$ in $\Omega = \{1,2,3,4,5\}$, the shaded regions show $A \cup B$, $A \cap B$, and $A \setminus B$, respectively. Each numeral's position records its membership; 5 belongs to the universal set but to neither A nor B.

EXERCISE 2.4

Please prove that $A \subseteq B$ if and only if $A \cup B = B$.

WORKED SOLUTION

First direction: suppose $A \subseteq B$. The definition of union gives $A \cup B \subseteq B$, while $B \subseteq A \cup B$ always holds. Therefore $A \cup B = B$.
Second direction: suppose $A \cup B = B$. Let $x \in A$. Then $x \in A \cup B$, so $x \in B$. Therefore

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EXERCISE 2.4 CONTINUED

$A \subseteq B$.

EXERCISE 2.5

Please prove that $A \subseteq B$ if and only if $A \cap B = A$.

WORKED SOLUTION

First direction: suppose $A \subseteq B$. If $x \in A \cap B$, then $x \in A$ and $x \in B$, so $A \cap B \subseteq A$. Conversely, if $x \in A$, then the inclusion gives $x \in B$. Thus $x \in A \cap B$, proving $A \subseteq A \cap B$. Hence $A \cap B = A$.

Second direction: suppose $A \cap B = A$. Let $x \in A$. Then $x \in A \cap B$, so $x \in B$. Therefore $A \subseteq B$.

2.3 Algebraic structure: fields

2.3.1 The field axioms

RECALL AND TOOLS

A field is a set F equipped with two operations, addition and multiplication. For all $a, b, c \in F$, the operations satisfy the following axioms. In particular, $0_{\mathbb{F}} \neq 1_{\mathbb{F}}$ is required:

- **Closure:** if $a, b \in F$, then $a +_{\mathbb{F}} b \in F$ and $a \times_{\mathbb{F}} b \in F$.
- **Commutativity:** $a +_{\mathbb{F}} b = b +_{\mathbb{F}} a$ and $a \times_{\mathbb{F}} b = b \times_{\mathbb{F}} a$.
- **Associativity:** $(a +_{\mathbb{F}} b) +_{\mathbb{F}} c = a +_{\mathbb{F}} (b +_{\mathbb{F}} c)$ and $(a \times_{\mathbb{F}} b) \times_{\mathbb{F}} c = a \times_{\mathbb{F}} (b \times_{\mathbb{F}} c)$.
- **Distributivity:** $a \times_{\mathbb{F}} (b +_{\mathbb{F}} c) = a \times_{\mathbb{F}} b +_{\mathbb{F}} a \times_{\mathbb{F}} c$ and $(a +_{\mathbb{F}} b) \times_{\mathbb{F}} c = a \times_{\mathbb{F}} c +_{\mathbb{F}} b \times_{\mathbb{F}} c$.
- **Identity elements:** there is $0_{\mathbb{F}} \in F$ such that $a +_{\mathbb{F}} 0_{\mathbb{F}} = a$ for every $a \in F$, and there is $1_{\mathbb{F}} \in F$ such that $a \times_{\mathbb{F}} 1_{\mathbb{F}} = a$ for every $a \in F$.
- **Additive and multiplicative inverses:** every $a \in F$ has an additive inverse $-a$ with $a +_{\mathbb{F}} (-a) = 0_{\mathbb{F}}$; every nonzero $a \in F$ has a multiplicative inverse a^{-1} with $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$.

EXAMPLE 2.1

- **Examples:** $\mathbb{Q}$ (rational numbers), $\mathbb{R}$ (real numbers), and $\mathbb{C}$ (complex numbers) are fields with their usual addition and multiplication.
- **Nonexamples:** $\mathbb{Z}$ and $\mathbb{N}$ are not fields, because most of their elements have no multiplicative inverse; for example, 2 has no inverse in $\mathbb{Z}$.

2.3.2 Notation

A general field is usually denoted by F (sometimes $\mathbb{F}$). When no particular field is intended, we write ``Let F be a field.''

Notation convention: when the field is clear from context, we omit its subscript from the operations and the elements $0, 1$.

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EXERCISE 2.6

Let $a \in F$ with $a \neq 0_{\mathbb{F}}$. Please prove that $(a^{-1})^{-1} = a$.

WORKED SOLUTION

Recall that every invertible field element has a unique multiplicative inverse, and in particular $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$. Put $x = (a^{-1})^{-1}$ and calculate:

$$(a^{-1}) \times_{\mathbb{F}} x = 1_{\mathbb{F}} \quad / \times_{\mathbb{F}} a$$

$$a \times_{\mathbb{F}} (a^{-1}) \times_{\mathbb{F}} x = a \times_{\mathbb{F}} 1_{\mathbb{F}}$$

$$1_{\mathbb{F}} \times_{\mathbb{F}} x = a \qquad (*)$$

Thus $x = a$, that is, $(a^{-1})^{-1} = a$.
Remark: step $(*)$ uses $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$.

EXERCISE 2.7

Let $a \in F$ with $a \neq 0_{\mathbb{F}}$. Please prove that the multiplicative inverse of a is unique.

WORKED SOLUTION

Suppose $b \in F$ satisfies $a \times_{\mathbb{F}} b = 1_{\mathbb{F}}$. Multiply on the left by a^{-1}:

$$a \times_{\mathbb{F}} b = 1_{\mathbb{F}} \quad / \times_{\mathbb{F}} a^{-1}$$

$$(a^{-1}) \times_{\mathbb{F}} a \times_{\mathbb{F}} b = a^{-1} \times_{\mathbb{F}} 1_{\mathbb{F}}$$

$$1_{\mathbb{F}} \times_{\mathbb{F}} b = a^{-1} \qquad (*)$$

Hence $b = a^{-1}$, so the inverse of a is unique. *Remark:* again, step $(*)$ uses $(a^{-1}) \times_{\mathbb{F}} a = 1_{\mathbb{F}}$.

2.4 Further exercises on fields

EXERCISE 2.8

Let a, b be elements of a field $\mathbb{F}$. Please prove that if $a \times_{\mathbb{F}} b = 0_{\mathbb{F}}$, then $a = 0_{\mathbb{F}}$ or $b = 0_{\mathbb{F}}$.

WORKED SOLUTION

We distinguish two cases according to b. **Case 1:** $b = 0_{\mathbb{F}}$, and the conclusion already holds. **Case 2:** $b \neq 0_{\mathbb{F}}$. Since $\mathbb{F}$ is a field, b has an inverse b^{-1}. Multiplying both sides by b^{-1} gives

$$a \times_{\mathbb{F}} b \times_{\mathbb{F}} b^{-1} = 0_{\mathbb{F}} \times_{\mathbb{F}} b^{-1} \Longrightarrow a \times_{\mathbb{F}} 1_{\mathbb{F}} = a = 0_{\mathbb{F}}$$

PROPOSITION 2.1

A field has no nontrivial zero divisors.

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EXERCISE 2.9

Let $\mathbb{F}$ be a field. Define a new structure $(F \times F, \oplus, \otimes)$ by

$$(a, b) \oplus (c, d) = (a +_{\mathbb{F}} c, b +_{\mathbb{F}} d),$$
$$(a, b) \otimes (c, d) = (a \times_{\mathbb{F}} c, b \times_{\mathbb{F}} d).$$

Is this new structure a field?

WORKED SOLUTION

Every element of this structure is a pair of elements of the given field, as with a point in a coordinate system.
The operation $\oplus$ adds the first and second components separately; multiplication $\otimes$ is defined in the same componentwise way.
We will show that the structure is not a field by finding two nonzero elements whose product is zero.
Suppose, for a contradiction, that $\mathbb{F} \times \mathbb{F}$ is a field. Its zero element is $(0_{\mathbb{F}}, 0_{\mathbb{F}})$, as follows from the field operations in each component. However,

$$\begin{aligned}(0_{\mathbb{F}}, 1_{\mathbb{F}}) \otimes (1_{\mathbb{F}}, 0_{\mathbb{F}}) &= (0_{\mathbb{F}} \times_{\mathbb{F}} 1_{\mathbb{F}}, 1_{\mathbb{F}} \times_{\mathbb{F}} 0_{\mathbb{F}}) \\ &= (0_{\mathbb{F}}, 0_{\mathbb{F}}).\end{aligned}$$

Neither factor is zero, contradicting the result of the preceding exercise.

2.5 Finite fields

NOTE

For a prime p, we write $\mathbb{Z}_p = \{0, \ldots, p-1\}$ and perform addition and multiplication modulo p. These integers represent residue classes. The same field is also denoted $\mathbb{F}_p$. Most finite-field examples here use these residue fields.

Definition and notation

DEFINITION 2.4

- A **finite field** is a field with finitely many elements. Write $|\mathbb{F}| = q$.
- Every finite field has $q = p^n$ elements, where p is prime and $n \geq 1$ is an integer. This is a structural theorem about finite fields.
- For a prime p, the set $\mathbb{Z}_p = \mathbb{Z}/p\mathbb{Z}$, with addition and multiplication modulo p, is a field. It is called the **prime field of characteristic** p.
- Addition and multiplication in $\mathbb{Z}_p$ are defined by

$$a +_{\mathbb{Z}_p} b := (a + b) \bmod p, \quad a \times_{\mathbb{Z}_p} b := (a \cdot b) \bmod p.$$

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Examples

EXAMPLE 2.2

- $\mathbb{Z}_2 = \{0, 1\}$, with $1 + 1 = 0$ and $1 \cdot 1 = 1$.
- $\mathbb{Z}_3 = \{0, 1, 2\}$, with addition and multiplication modulo 3.
- $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$, with operations modulo 5.

Basic properties

PROPOSITION 2.2

In every finite field $\mathbb{F}_q$, the set $\mathbb{F}_q^\times$ of nonzero elements is closed under multiplication, and each of its elements has a unique inverse.

PROPOSITION 2.3

In every finite field $\mathbb{F}_q$, each $a \in \mathbb{F}_q$ satisfies

$$a^q = a.$$

PROPOSITION 2.4

Every nonzero a in a finite field $\mathbb{F}_q$ satisfies $a^{q-1} = 1$.

Explanation: Closure follows because a field has no zero divisors. The existence and uniqueness of inverses follow from the field axioms and the proof above. To prove both power identities, let $a \neq 0$ and put

$$P = \prod_{x \in \mathbb{F}_q^\times} x \neq 0.$$

Multiplication by a permutes the nonzero elements: the map $x \mapsto ax$ is invertible, with inverse $x \mapsto a^{-1}x$. Thus

$$P = \prod_{x \in \mathbb{F}_q^\times} (ax) = a^{q-1}P.$$

Since $P \neq 0$, cancellation is permitted and gives $a^{q-1} = 1$, hence $a^q = a$. For $a = 0$, the identity $0^q = 0$ is immediate.

EXAMPLE 2.3

Calculate all inverses in $\mathbb{Z}_7 = \mathbb{Z}/7\mathbb{Z}$, for $1 \leq a \leq 6$:

a	1	2	3	4	5	6
a^{-1}	1	4	5	2	3	6

For example, $2 \cdot 4 = 8 \equiv 1$, $3 \cdot 5 = 15 \equiv 1$, and $4 \cdot 2 \equiv 1$, and similarly for the remaining entries.

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EXERCISE 2.10

Please show that, in every finite field $\mathbb{F}_q$,

$$(a-b)^q = a^q - b^q.$$

WORKED SOLUTION

Use the identity $x^q = x$, proved for every $x \in \mathbb{F}_q$. In particular, it applies to $a-b$, so

$$(a-b)^q = a - b = a^q - b^q.$$

This proves the claim directly, without calculating binomial coefficients.

EXERCISE 2.11

Work in $\mathbb{Z}_{13} = \mathbb{Z}/13\mathbb{Z}$.

1. Please find the inverses of $2, 3, 4, 5, 6$.
2. Please evaluate the following expression, checking that every denominator is nonzero:

$$E = (5^{-1} + 2)(4 - 3^{-1})^{-1} - 1.$$

WORKED SOLUTION

All arithmetic is modulo 13. Check the products

$$2 \cdot 7 = 14 \equiv 1, \quad 3 \cdot 9 = 27 \equiv 1, \quad 4 \cdot 10 = 40 \equiv 1,$$

$$5 \cdot 8 = 40 \equiv 1, \qquad 6 \cdot 11 = 66 \equiv 1.$$

The inverses, in the stated order, are therefore $7, 9, 10, 8, 11$. Now $5^{-1} + 2 = 8 + 2 = 10$, whereas

$$4 - 3^{-1} = 4 - 9 = -5 \equiv 8 \neq 0.$$

The inverse of 8 is 5, because $8 \cdot 5 = 40 \equiv 1$. Consequently,

$$E \equiv 10 \cdot 5 - 1 = 49 \equiv 10 \pmod{13}.$$

Every inverse appearing in the expression is defined, and the answer is the residue class 10.

EXERCISE 2.12

Please show that for every nonzero a in a finite field $\mathbb{F}_q$, there is $n \in \mathbb{N}$ with $a^n = 1$.

WORKED SOLUTION

The set $\mathbb{F}_q^\times$ contains exactly $q-1$ elements and is closed under multiplication. Put $m = q-1$ and consider the list

$$a, a^2, a^3, \ldots, a^{m+1}.$$

It has $m+1$ entries but only m possible distinct values, so two entries are equal, say $a^k = a^l$ with $k > l$. Multiplying by a^{-l} gives $a^{k-l} = 1$. Thus every nonzero element has a positive

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EXERCISE 2.12 CONTINUED

integer power equal to the multiplicative identity.

2.6 Calculations and applications in finite fields

EXERCISE 2.13

A system stores two information bits $a, b \in \mathbb{Z}_2$ and adds a check bit c such that $a+b+c=0$. Please find c. If one of the three bits is flipped, does the check detect it? What happens if exactly two bits are flipped?

WORKED SOLUTION

In $\mathbb{Z}_2$, $-x = x$, so $c = a+b$. For example, if $a=b=1$, then $c=0$. Let $e_1, e_2, e_3 \in \mathbb{Z}_2$ denote the errors, where $e_j = 1$ means that bit j is flipped. The sum of the received bits is

$$(a+e_1)+(b+e_2)+(c+e_3) = (a+b+c)+(e_1+e_2+e_3) = e_1+e_2+e_3.$$

For a single flip, this sum is 1, so an error is detected. For two flips, it is $1+1=0$, so that error is undetected. The check detects any single flip but does not identify which bit was flipped.

Additional practice

EXERCISE 2.14 FOUNDATION

In the field $\mathbb{Z}_7$, define

$$A = \{x \in \mathbb{Z}_7 : 2x+3=1\}, \qquad B = \{x \in \mathbb{Z}_7 : x^2 = 1\}.$$

Please find A, B, $A \cup B$, $A \cap B$, and please determine whether $A \subseteq B$.

WORKED SOLUTION

All arithmetic is modulo 7. The first equation is equivalent to $2x=5$. The inverse of 2 is 4, so $x = 4 \cdot 5 = 20 = 6$ in the field, giving $A = \{6\}$. In the second equation, $x^2-1 = (x-1)(x+1) = 0$. Since a field has no nontrivial zero divisors, necessarily $x=1$ or $x=-1=6$. Both values do satisfy the equation, so $B = \{1,6\}$. Hence

$$A \subseteq B, \qquad A \cup B = \{1,6\} = B, \qquad A \cap B = \{6\} = A.$$

EXERCISE 2.15 CHALLENGE

Let F be a finite field with $q > 2$ elements. Please prove that

$$\sum_{x \in F} x = 0, \qquad \sum_{x \in F^\times} x^{-1} = 0.$$

Please check both statements in the field with two elements. Do not assume an ordering on F.

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EXERCISE 2.15 CHALLENGE CONTINUED

WORKED SOLUTION

Put $S = \sum_{x \in F} x$. Since $q > 2$, choose $c \in F \setminus \{0, 1\}$. The map $x \mapsto cx$ is bijective, with inverse $x \mapsto c^{-1}x$. It therefore only rearranges the summands, and

$$cS = \sum_{x \in F} cx = \sum_{x \in F} x = S.$$

Thus $(c-1)S = 0$. Since $c - 1 \neq 0$, field cancellation gives $S = 0$. Removing the summand 0 does not change the sum, so $\sum_{x \in F^\times} x = 0$. The map $x \mapsto x^{-1}$ also permutes $F^\times$, because applying it twice returns x. Therefore

$$\sum_{x \in F^\times} x^{-1} = \sum_{x \in F^\times} x = 0.$$

In $\mathbb{Z}_2$, we have $0 + 1 = 1 \neq 0$, and the sum of the nonzero inverses is $1^{-1} = 1 \neq 0$. Indeed, no element c distinct from $0, 1$ exists there, so the choice used in the proof is unavailable.

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EXERCISE 2.16 CHALLENGE

Let F be a field satisfying $x^3 = x$ for every $x \in F$. Please prove that F has either two or three elements. Please write its operations in each case and please identify them with those of $\mathbb{Z}_2$ or $\mathbb{Z}_3$.

WORKED SOLUTION

For $x \neq 0$, cancellation gives $x^2 = 1$, hence $(x-1)(x+1) = 0$. A field has no zero divisors, so $x = 1$ or $x = -1$. Thus $F \subseteq \{0, 1, -1\}$. If $1 = -1$, then $F = \{0, 1\}$ and $1 + 1 = 0$. Together with multiplication by zero and $1 \cdot 1 = 1$, these are the operations of $\mathbb{Z}_2$. If $1 \neq -1$, all three elements are distinct and present. The sum $1 + 1$ cannot be 0, which would give $1 = -1$, or 1, which would give $1 = 0$ by cancellation. Therefore $1 + 1 = -1$. Writing $2 = -1$, we obtain

$$1 + 2 = 0, \qquad 2 + 2 = 1, \qquad 2 \cdot 2 = 1.$$

These, together with the zero and identity rules, give exactly $\mathbb{Z}_3$. Conversely, direct substitution of $0, 1$ in $\mathbb{Z}_2$, and $0, 1, -1$ in $\mathbb{Z}_3$, verifies $x^3 = x$.

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EXERCISE 2.17 CHALLENGE

Let F be a finite field. Please prove

$$\prod_{x \in F \setminus \{0\}} x = -1.$$

Here $\prod$ multiplies all nonzero elements. Please also prove that, for every $b \in F$,

$$\prod_{a \in F \setminus \{b\}} (b - a) = -1.$$

WORKED SOLUTION

Pair each nonzero x with x^{-1}, except when the two are equal. Each pair contributes 1 to the product. The unpaired elements satisfy $x = x^{-1}$, equivalently

$$x^2 = 1, \qquad (x-1)(x+1) = 0.$$

Since a field has no zero divisors, these are exactly 1 and -1. If they are distinct, their product is -1. If $1 = -1$, only the element 1 remains unpaired, and its value is again -1. This proves the first identity in every characteristic. For fixed b, the values $b - a$ as a ranges over $F \setminus \{b\}$ run through every nonzero field element exactly once. The second product is therefore the same product in a different order.

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